Worksheet 49 · Vectors IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Add, subtract and multiply vectors • Write vector paths in terms of given vectors • Use vectors to prove points are collinear or lines parallel		
Grade: / 66	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

$\overrightarrow{AB}$ → $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

Magnitude of $\begin{pmatrix} x \\ y \end{pmatrix}$ → $\left|\begin{pmatrix} x \\ y \end{pmatrix}\right| = \sqrt{x^2+y^2}$

Parallel vectors → Multiples of one another $\overrightarrow{AB} = 2 \times \overrightarrow{CD}$

Collinear → A, B, C lie on a straight line

↳ Any two of $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{BC}$ are parallel.

Questions *Vectors*

Question 1 (5 marks)

$ABCD$ is a trapezium.

$\overrightarrow{DC} = 3\overrightarrow{AB}$

$\overrightarrow{DA} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ $\overrightarrow{DB} = \begin{pmatrix} -1 \\ 7 \end{pmatrix}$

Find the exact magnitude of $\overrightarrow{BC}$

Question 2 (5 marks)

OAB is a triangle.

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OB} = \mathbf{b}$

C is the midpoint of OA.
D is the point on AB such that $AD : DB = 3 : 1$
E is the point such that $\overrightarrow{OB} = 2\overrightarrow{BE}$

Using a vector method, prove that the points C, D and E lie on the same straight line.

Question 3 (5 marks)

The diagram shows trapezium $OACB$.

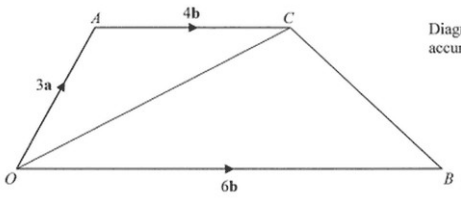


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 3\mathbf{a}$ $\overrightarrow{OB} = 6\mathbf{b}$ $\overrightarrow{AC} = 4\mathbf{b}$

N is the point on OC such that ANB is a straight line.

Find $\overrightarrow{ON}$ as a simplified expression in terms of **a** and **b**.

Question 4 (2 marks)

Here are two vectors.

$\overrightarrow{AB} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}$ $\overrightarrow{CB} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$

Find, as a column vector, $\overrightarrow{AC}$

Question 5 (5 marks)

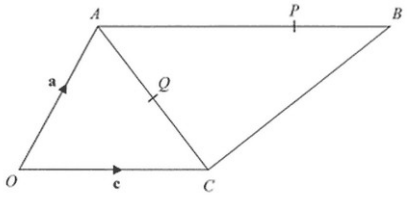


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OC} = \mathbf{c}$ $\overrightarrow{AB} = 2\mathbf{c}$

P is the point on AB such that $AP : PB = 3 : 1$
Q is the point on AC such that OQP is a straight line.

Use a vector method to find $AQ : QC$
Show your working clearly.

$AQ : QC =$ .....

Question 6 (5 marks)

$ABCDEF$ and $GHIJKL$ are regular hexagons each with centre O.

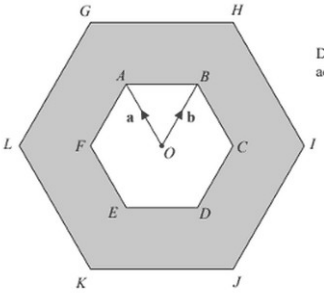


Diagram **NOT** accurately drawn

$GHIJKL$ is an enlargement of $ABCDEF$, with centre O and scale factor 2

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OB} = \mathbf{b}$

(a) Write the following vectors, in terms of **a** and **b**.
Simplify your answers.

(i) $\overrightarrow{AB}$

.....
(1)

(ii) $\overrightarrow{KI}$

(2)

(iii) $\overrightarrow{LD}$

.....
(2)

Question 7 (8 marks)

The triangle OAB has an area of 5 cm^2

(b) Calculate the area of the shaded region.

..... cm^2
(3)

Question 8 (3 marks)

$ABCD$ is a parallelogram.

$$\overrightarrow{AB} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \overrightarrow{AC} = \begin{pmatrix} 9 \\ 4 \end{pmatrix}$$

Find the magnitude of $\overrightarrow{BC}$

.....

Question 9 (6 marks)

OAB is a triangle.

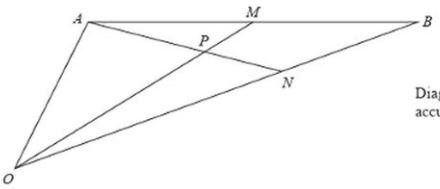


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 2\mathbf{a}$ and $\overrightarrow{OB} = 2\mathbf{b}$

M is the midpoint of AB.
N is the point on OB such that $ON : NB = 2 : 1$

P is the point on AN such that OPM is a straight line.

Use a vector method to find $OP : PM$
Show your working clearly.

.....

Question 10 (5 marks)

The diagram shows triangle OAB

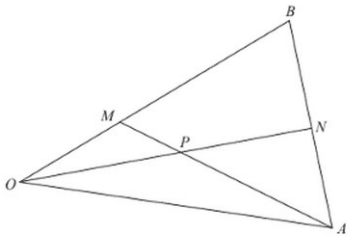


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 8\mathbf{a}$ $\overrightarrow{OB} = 6\mathbf{b}$

M is the point on OB such that $OM : MB = 1 : 2$
N is the midpoint of AB
P is the point of intersection of ON and AM

Using a vector method, find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**
Show your working clearly.

$\overrightarrow{OP} =$ .....

Question 11 (4 marks)

$ABCD$ is a parallelogram and ADM is a straight line.

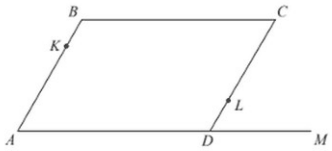


Diagram **NOT** accurately drawn

$\overrightarrow{AB} = \mathbf{a}$ $\overrightarrow{BC} = \mathbf{b}$ $\overrightarrow{DM} = \frac{1}{2}\mathbf{b}$

K is the point on AB such that $AK : AB = \lambda : 1$
L is the point on CD such that $CL : CD = \mu : 1$
KLM is a straight line.

Given that $\lambda : \mu = 1 : 2$

use a vector method to find the value of λ and the value of μ

$\lambda =$ .....

$\mu =$ .....

Question 12 (5 marks)

$OACB$ is a trapezium.

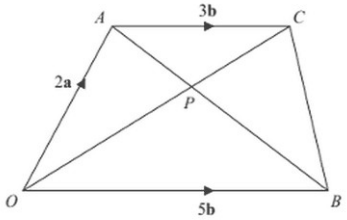


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 2\mathbf{a}$ $\overrightarrow{OB} = 5\mathbf{b}$ $\overrightarrow{AC} = 3\mathbf{b}$

The diagonals, OC and AB, of the trapezium intersect at the point P.

Find and simplify an expression, in terms of **a** and **b**, for $\overrightarrow{OP}$
Show your working clearly.

$\overrightarrow{OP} =$ .....

Question 13 (3 marks)

$ABCDEF$ is a regular hexagon.

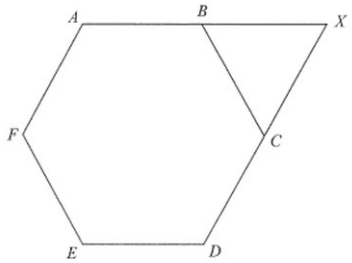


Diagram **NOT** accurately drawn

ABX and DCX are straight lines.

$\overrightarrow{AB} = \mathbf{a}$ $\overrightarrow{BC} = \mathbf{b}$

Find $\overrightarrow{EX}$ in terms of **a** and **b**.
Give your answer in its simplest form.

Question 14 (5 marks)

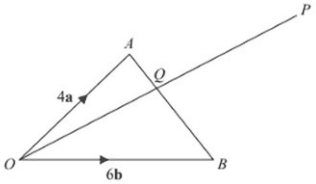


Diagram **NOT** accurately drawn

OAB is a triangle.
Q is the point on AB such that OQP is a straight line.

$\overrightarrow{OA} = 4\mathbf{a}$ $\overrightarrow{OB} = 6\mathbf{b}$ $\overrightarrow{AP} = 2\mathbf{a} + 8\mathbf{b}$

Using a vector method, find the ratio $AQ : QB$

$AQ : QB =$ .....

(Total for Question 21 is 5 marks)